ELEC70039
EE9-AML3-04
EE9-CS7-21
EE9-SO22

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

WAVELETS, REPRESENTATION LEARNING AND THEIR APPLICATIONS

Thursday, 15 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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EXAMINATIONS 2025

WAVELETS AND APPLICATIONS

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Special Information for the Invigilators: NONE

Information for Candidates:

Sub-sampling by an integer N

$$x_{\downarrow N}[n] \Longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

Dual Basis:
Given a basis $\{\varphi_i(t)\}_{i\in\mathbb{Z}}$, the dual basis is given by the set of elements $\{\tilde{\varphi}_i(t)\}_{i\in\mathbb{Z}}$ satisfying:

$$\langle \varphi_i(t), \tilde{\varphi}_j(t)\rangle = \delta_{i,j}.$$

The Questions

1. (a) Consider the multi-rate system shown in Figure 1a, where $G_0(z)$ is an ideal low-pass filter with cutoff frequency $\pi/2$ and $G_1(z)$ is an ideal high-pass filter with cutoff frequency $\pi/2$. The two filters are shown in Figure 1b.
Sketch and dimension the four spectra $Y_1(e^{j\omega})$, $Y_2(e^{j\omega})$, $Y_3(e^{j\omega})$, $Y_4(e^{j\omega})$ assuming that $x[n]$ has the spectrum shown in Figure 1c.

[5]

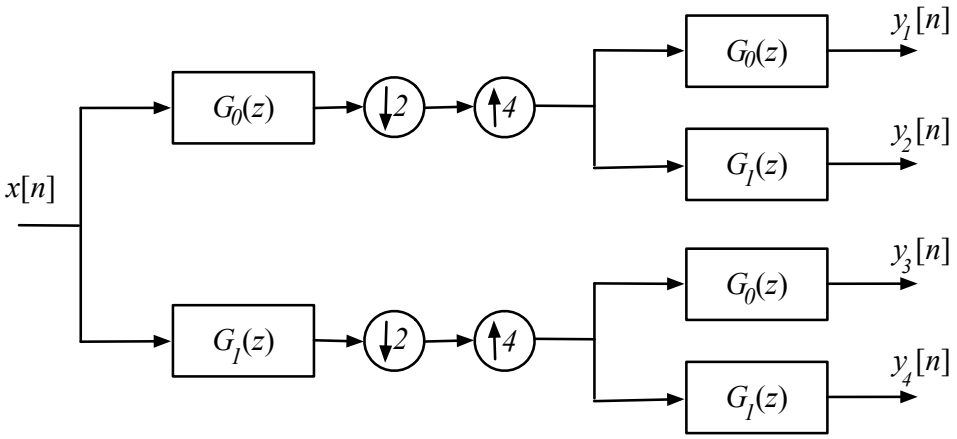


Figure 1a: The multi-rate system.

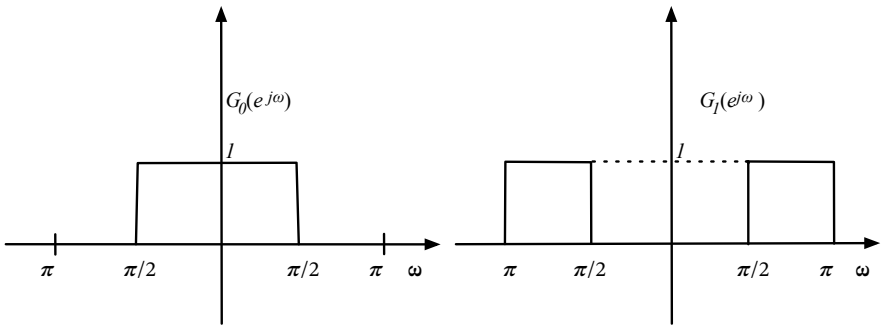


Figure 1b: The ideal low-pass and high-pass filters $G_0(z)$, $G_1(z)$.

Question continues on next page.

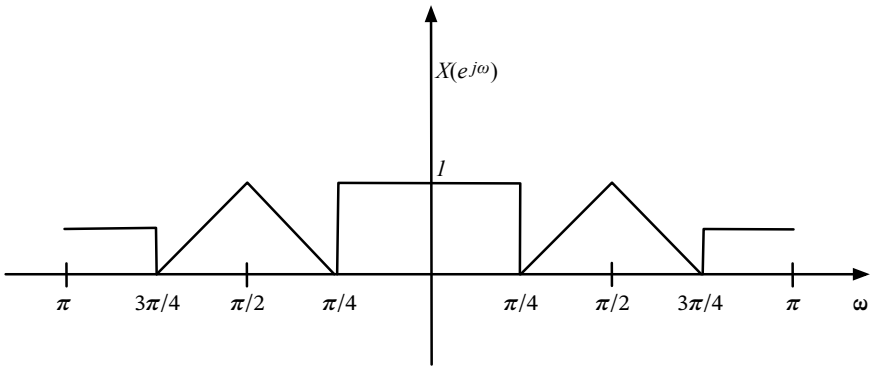


Figure 1c: Spectrum of $x[n]$.

(b) Now assume $G_0(z) = (1 + z^4)$ and $G_1(z) = (1 - z^4)$.

i. Draw the equivalent single-level four-channel filter-bank, clearly specifying the z-transform of the equivalent filters.

[5]

ii. Assume that $x[n] = 1$ and ignore any boundary effect. Which of the signals $y_1[n], y_2[n], y_3[n]$ and $y_4[n]$ is nonzero? (Justify your answer).

[5]

iii. Assume now that $x[n] = n$ and ignore again any boundary effect. Which of the signals $y_1[n], y_2[n], y_3[n]$ and $y_4[n]$ is nonzero? (Justify your answer).

[5]

iv. Find the z-transform of $G_1(z)$ that would lead to $y_1[n], y_2[n], y_3[n]$ being nonzero and $y_4[n] = 0$ when $x[n] = n^2$. Ignore again any boundary effect and assume $G_0(z) = (1 + z^4)$. Please note that the solution is not unique.

[5]

2. Consider the two-channel filter bank of Figure 2a.

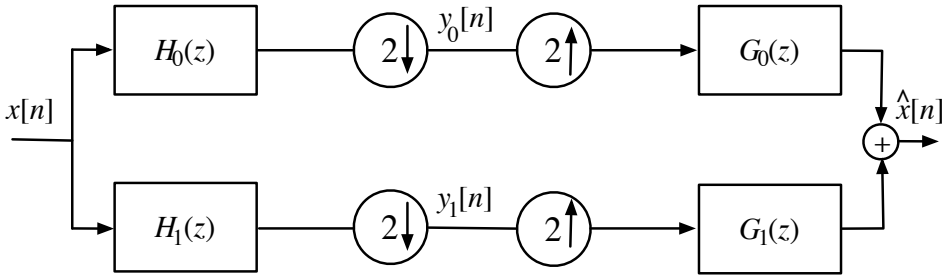


Figure 2a: Two-channel filter bank.

(a) Express $\hat{X}(z)$ as a function of $X(z)$ and the filters. Then derive the two perfect reconstruction (PR) conditions the filters have to satisfy. [5]

(b) Assume

$$P(z) = G_0(z)H_0(z) = \left(\frac{z^3}{2} + 1 + \frac{z^{-3}}{2}\right).$$

i. Design an orthogonal filter bank assuming:

$$H_0(z) = \frac{1}{\sqrt{2}}(1 + z^3).$$

[7]

ii. Now assume $H_0(z) = (z - 1 + z^{-1})$, build the corresponding bio-orthogonal filter bank. [7]

(c) Consider the two-channel filter bank of Figure 2a without downsamplers and upsamplers. Such a filter bank is called a *Nonsubsampled* filter bank.

i. Choose $\{H_0(z), H_1(z), G_0(z), G_1(z)\}$ as in a two-channel orthogonal filter bank. What is $\hat{x}[n]$ as a function of $x[n]$ and the filters? [3]

ii. Assume that $H_0(z) = (z+2)$ and that $G_0(z) = H_0(z^{-1})$. Calculate $\{H_1(z), G_1(z)\}$ in order to achieve perfect reconstruction. You want $\{H_1(z), G_1(z)\}$ to be real valued and you want to find at least two alternative solutions. [3]

3. Consider a set of linearly independent functions $\{\varphi_i\}_{i=1,2,..,N}$ that covers the sub-space V, that is, $V = span\{\varphi_i\}_{i=1,2,...N}$. Consider a function $f(t) \in L_2(\mathbb{R})$, the orthogonal projection of $f(t)$ onto V is:

$$\hat{f}(t) = \sum_{i=1}^{N} c_i \varphi_i(t),$$

with $c_i = \langle f(t), \tilde{\varphi}_i(t) \rangle$, $i = 1, 2, ...N$. Here $\{\tilde{\varphi}_i\}_{i=1,2,..,N}$ is the dual basis of $\{\varphi_i\}_{i=1,2,..,N}$. Based on the above formula and assuming

$$f(t) = \pi \sin(2\pi t) \qquad t \in [0, 1/2],$$

(a) compute the coefficients c_i of the orthogonal projection of $f(t)$ onto the space spanned by $\varphi(t), \psi(t), \sqrt{2}\psi(2t), \sqrt{2}\psi(2t-1)$ with

$$\varphi(t) = \begin{cases} 1 & \text{for} \quad 0 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\psi(t) = \begin{cases} 1 & \text{for} \quad 0 \leq t < 1/2 \\ -1 & \text{for} \quad 1/2 \leq t < 1 \\ 0 & \text{otherwise,} \end{cases}$$

[5]

(b) sketch and dimension the function $\hat{f}(t)$,

[5]

(c) compute the energy of the error function $\epsilon(t) = f(t) - \hat{f}(t)$,

[5]

(d) verify that $\|f(t)\|^2 = \|\hat{f}(t)\|^2 + \|\epsilon(t)\|^2$, where $\|x(t)\|^2 = \int_{-\infty}^{\infty} x^2(t)dt$.

[5]

(e) Now assume $f(t) = \varphi(t)$, find the expression for the wavelet coefficients:

$$a_{m,n} = \langle f(t), \psi_{m,n}(t) \rangle,$$

where $\psi_{m,n}(t) = 2^{-m/2}\psi(2^{-m}t - n)$.

[5]

4. (a) Show that the two-scale equation

$$\varphi(t) = \sqrt{2} \sum_{n=-\infty}^{\infty} g_0[n]\varphi(2t-n)$$

can be written in the Fourier domain as follows:

$$\hat{\varphi}(\omega) = \frac{1}{\sqrt{2}} G_0(e^{j\omega/2})\hat{\varphi}(\omega/2),$$

where $\hat{\varphi}(\omega)$ is the Fourier transform of $\varphi(t)$ and $G_0(e^{j\omega})$ is the discrete-time Fourier transform of $g_0[n]$. [5]

(b) Assume the function $f(t)$ satisfies $\int_{-\infty}^{\infty} |\hat{f}(\omega)|d\omega < +\infty$, where $\hat{f}(\omega)$ is the Fourier transform of $f(t)$. Show that $f(t)$ is a continuous function. [5]

(c) Suppose you are given a two-channel FIR filter bank with real coefficients and synthesis lowpass filter

$$g_0[n] = \frac{1}{8\sqrt{2}}(\delta_n + 4\delta_{n-1} + 6\delta_{n-2} + 4\delta_{n-3} + \delta_{n-4}).$$

Consider the equivalent filter

$$G_0^{(i)}(z) = \prod_{k=0}^{i-1} G_0(z^{2^k})$$

obtained by iterating the filter bank decomposition i times. Consider the function

$$\varphi^{(i)}(t) = 2^{i/2} g_0^{(i)}[n], \qquad n/2^i \le t < (n+1)/2^i.$$

i. Can you say anything about the convergence of $\lim_{i\to\infty} \varphi^{(i)}(t)$? [5]

ii. We know that, in the case of convergence, $\varphi(t)$ is a valid scaling function. Can you say anything about continuity of this function? [5]

iii. State the number of vanishing moments of the analysis wavelet function obtained from $\varphi(t)$. [5]